RIVER VALLEY HIGH SCHOOL
JC2 PRELIMINARY EXAMINATIONS

H2 PHYSICS 9478
PAPER 2
17 SEP 2026

2 HOURS

CANDIDATE NAME									
CENTRE NUMBER	S					**INDEX NUMBER**				
CLASS	2	5	J						

INSTRUCTIONS TO CANDIDATES

DO NOT OPEN THIS BOOKLET UNTIL YOU ARE TOLD TO DO SO.

Read these notes carefully.
Write your name, centre number, index number and class in the spaces at the top of this page.

Write in dark blue or black pen on both sides of the paper.
You may use an HB pencil for any diagrams or graphs.
Do not use staples, paper clips, glue or correction fluid.

The use of an approved scientific calculator is expected where appropriate.

Candidates answer on the Question Paper.
No Additional Materials are required.

*Answer **all** questions*.

The number of marks is given in brackets [ ] at the end of each question or part question.

FOR EXAMINERS' USE	
1	**/10**
2	**/6**
3	**/8**
4	**/12**
5	**/7**
6	**/8**
7	**/12**
8	**/12**
Deduction	
TOTAL	**/ 75**

This document consists of **24** printed pages.

Data

speed of light in free space	c	=	3.00×10^{8} m s^{-1}
permeability of free space	μ_0	=	$4\pi \times 10^{-7}$ H m^{-1}
permittivity of free space	ε_0	=	8.85×10^{-12} F m^{-1}
	$\left(\frac{1}{4\pi\varepsilon_0}\right)$	=	8.99×10^{9} m F^{-1})
elementary charge	e	=	1.60×10^{-19} C
Planck constant	h	=	6.63×10^{-34} J s
unified atomic mass constant	u	=	1.66×10^{-27} kg
rest mass of electron	m_e	=	9.11×10^{-31} kg
rest mass of proton	m_p	=	1.67×10^{-27} kg
molar gas constant	R	=	8.31 J K^{-1} mol^{-1}
Avogadro constant	N_A	=	6.02×10^{23} mol^{-1}
Boltzmann constant	k	=	1.38×10^{-23} J K^{-1}
gravitational constant	G	=	6.67×10^{-11} N m^2 kg^{-2}
acceleration of free fall	g	=	9.81 m s^{-2}

Formulae

uniformly accelerated motion	$s = ut + \frac{1}{2}at^2$
	$v^2 = u^2 + 2as$
work done on / by a gas	$W = p\Delta V$
pressure	$p = \frac{F}{A}$
gravitational potential	$\phi = -\frac{GM}{r}$
temperature	T / K = T / °C + 273.15
pressure of an ideal gas	$p = \frac{1}{3}\frac{Nm}{V}\langle c^2 \rangle$

mean translational kinetic energy of an ideal gas particle	$E = \frac{3}{2}kT$
displacement of particle in s.h.m.	$x = x_0 \sin \omega t$
velocity of particle in s.h.m.	$v = v_0 \cos \omega t = \pm\omega\sqrt{(x_0^2 - x^2)}$
electric current	$I = nAvq$
resistors in series	$R = R_1 + R_2 + \ldots$
resistors in parallel	$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} + \ldots$
capacitors in series	$\frac{1}{C} = \frac{1}{C_1} + \frac{1}{C_2} + \ldots$
capacitors in parallel	$C = C_1 + C_2 + \ldots$
energy in a capacitor	$U = \frac{1}{2}QV = \frac{1}{2}\frac{Q^2}{C} = \frac{1}{2}CV^2$
charging a capacitor	$Q = Q_0\left[1 - e^{-\frac{t}{\tau}}\right]$
discharging a capacitor	$Q = Q_0 e^{-\frac{t}{\tau}}$
RC time constant	$\tau = RC$
electric potential	$V = \frac{Q}{4\pi\varepsilon_0 r}$
alternating current / voltage	$x = x_0 \sin \omega t$
magnetic flux density due to a long straight wire	$B = \frac{\mu_0 I}{2\pi d}$
magnetic flux density due to a flat circular coil	$B = \mu_0 \ni \frac{¿}{2r} ¿$
magnetic flux density due to a long solenoid	$B = \mu_0 \ni ¿ ¿$
energy states for quantum particle in a box	$E_n = \frac{h^2}{8mL^2}n^2$
radioactive decay	$x = x_0 e^{-\lambda t}$
radioactive decay constant	$\lambda = \frac{\ln 2}{t_{\frac{1}{2}}}$

Answer **all** the questions in the spaces provided.

1 A small block is released from rest on a frictionless surface inclined at 35° to the horizontal and slides a certain distance before reaching points A and B, as shown in Fig. 1.1.

The block was released at time $t = 0$, passing point A at time t_1 and point B at time t_2.

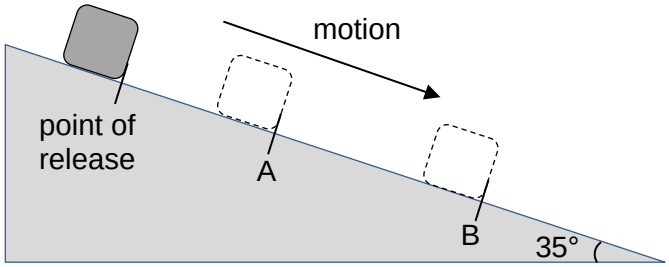


Fig. 1.1

(a) **(i)** Show that the acceleration of the block sliding down the frictionless surface is 5.6 m s^{-2}.

[1]

(ii) Sketch, in Fig. 1.2, the variation of the block's velocity v with time t, from $t = 0$ until the block reaches B. Mark the times t_1 and t_2 on the graph. No calculations are required.

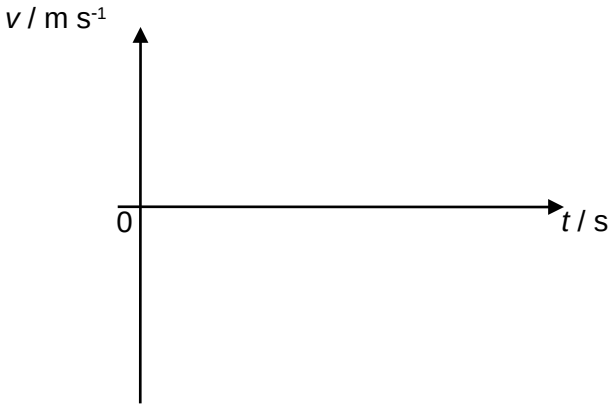


Fig. 1.2

[1]

(iii) The block takes 0.30 s to travel from A to B and reaches B with a velocity of 2.0 m s^{-1}. Hence or otherwise, determine the distance between A and B.

distance = …… m [2]

(iv) With reference to the definition of power, explain why the rate of increase of the block's kinetic energy with respect to time increases as the block moves down the incline.

……

……

……

…… [2]

(b) The block is now fitted with a large piece of cardboard, with its surface facing the direction of motion, and is released from rest at the same position on the frictionless 35° incline. It accelerates down the slope and eventually reaches terminal velocity due to air resistance.

(i) Explain why the block eventually reaches terminal velocity.

……

……

……

…… [2]

(ii) Sketch, in Fig. 1.3, a graph showing the variation of the block's acceleration *a* with time *t* for the block fitted with the piece of cardboard.

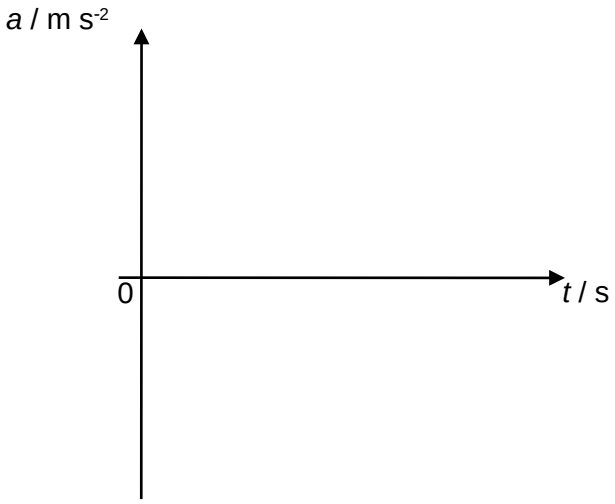


Fig. 1.3

[1]

(iii) Sketch, in Fig. 1.4, a graph showing the variation of the block's displacement *s* with time *t* for the block fitted with the piece of cardboard.

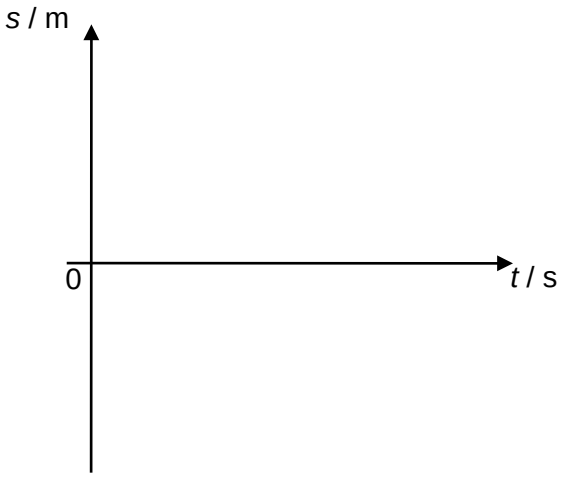


Fig. 1.4

[1]

2 A particle enters the region between two horizontal metal plates in a vacuum, as shown in Fig. 2.1. While the particle is between the plates, it experiences a constant upward acceleration of 24 m s^{-2}.

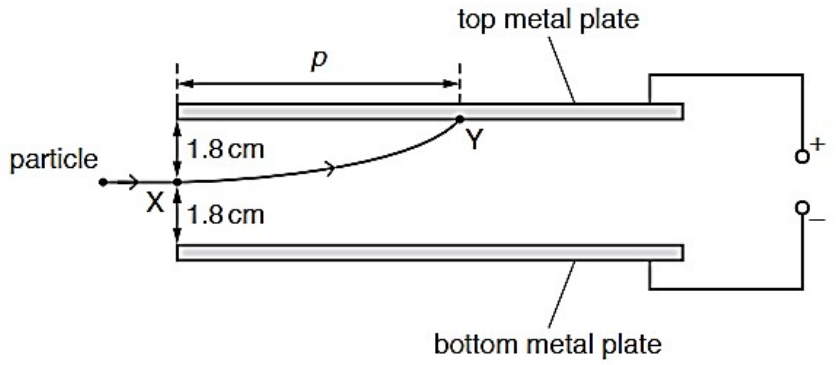


Fig. 2.1

The particle enters the electric field at point X with a horizontal speed of 0.75 m s^{-1} along a line midway between the plates.

The separation of the plates is 3.6 cm.

(a) Determine the time *t* for the particle to strike the upper plate.

t = ... s [2]

(b) Hence determine the minimum length *p* of the plates for the particle to just strike the upper plate.

p = ... cm [1]

(c) The plates have length 4.0 cm. The voltage is adjusted such that the net force acting on the particle remains unchanged. Determine the minimum separation between the plates such that the particle does not strike either plate.

minimum separation = …… cm [3]

3 A spring is used to join two frictionless trolleys A and B of mass M_1 and M_2 respectively, where $M_1 > M_2$ as shown in Fig. 3.1.

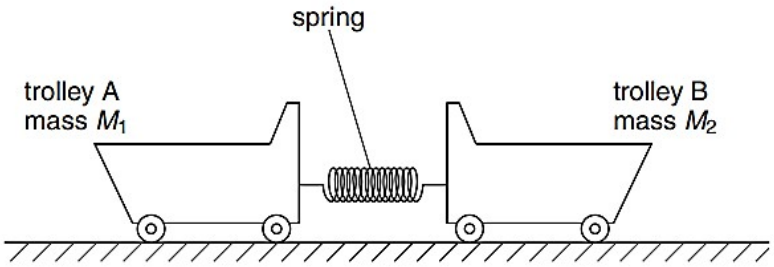


Fig. 3.1

The trolleys rest on a horizontal frictionless surface and are held apart so that the spring is extended.

The trolleys are then released.

(a) During the motion, the magnitude of the force exerted by the spring on each trolley varies with time.

With reference to Newton's laws of motion, explain why the magnitude of the impulse delivered to trolley A is equal to that delivered to trolley B.

……

……

……

……

…… [3]

(b) Explain why the centre of gravity of the two-trolley system remains at rest throughout the motion.

……

……

……

…… [2]

(c) Determine an expression for the ratio of the kinetic energy of trolley A to that of trolley B in terms of the masses of the two trolleys.

[3]

4 **(a)** A ball is held between two fixed points A and B by means of two stretched springs as shown in Fig. 4.1.

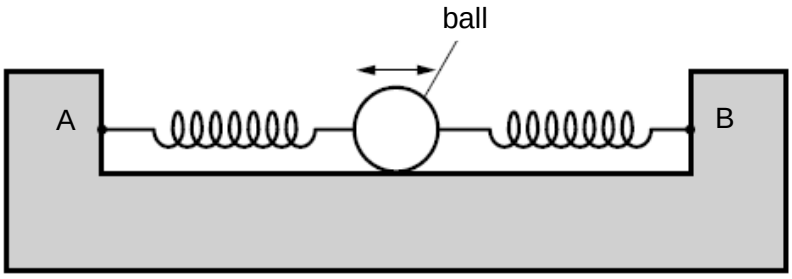


Fig. 4.1

The ball is free to oscillate horizontally along the line AB on the smooth plane.

The variation of the acceleration *a* of the ball with its displacement *x* from its equilibrium position is shown in Fig. 4.2.

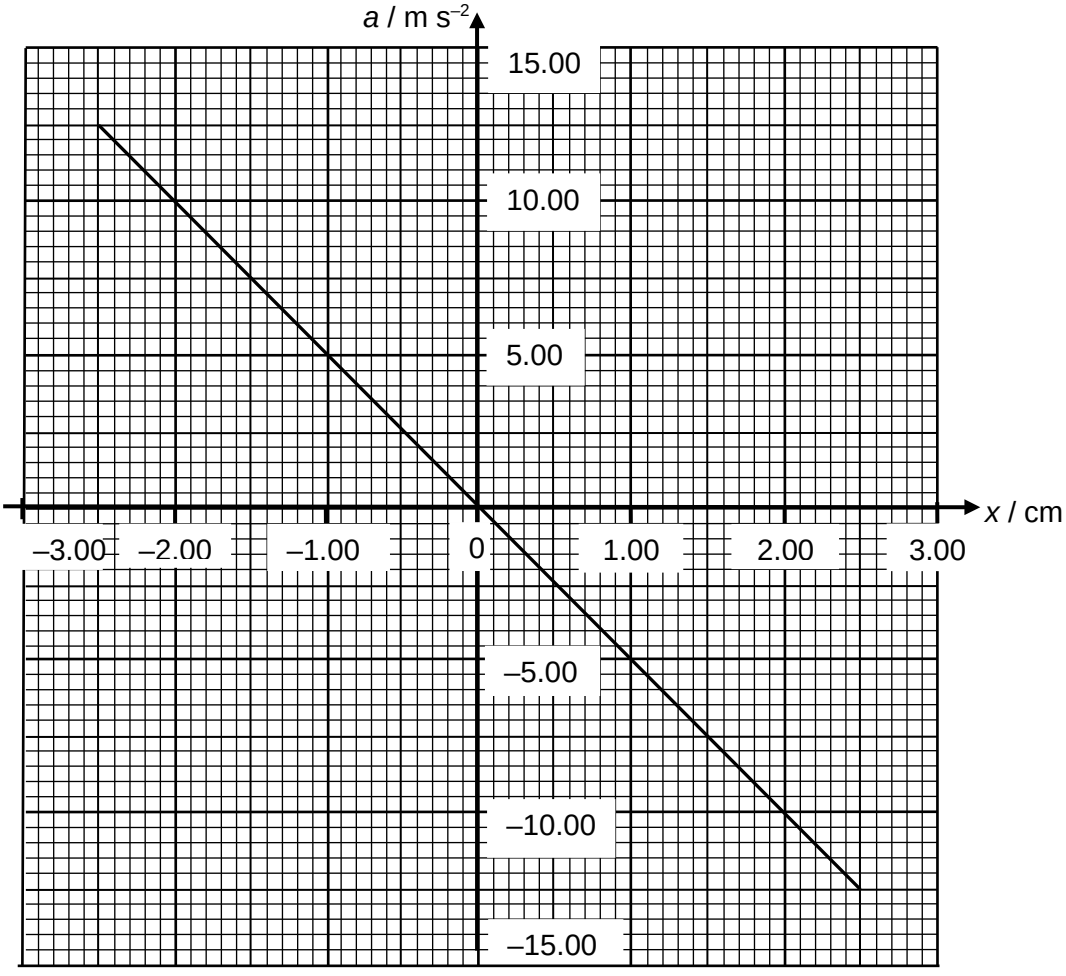


Fig. 4.2

(i) State and explain the features of Fig. 4.2 which indicate that the motion of the ball is simple harmonic.

……..

……..

……..

…… [2]

(ii) On Fig. 4.3, sketch the labelled velocity-displacement graph of the simple harmonic motion which was illustrated in Fig. 4.2. Intersection points on the horizontal and vertical axis need to be indicated in this figure.

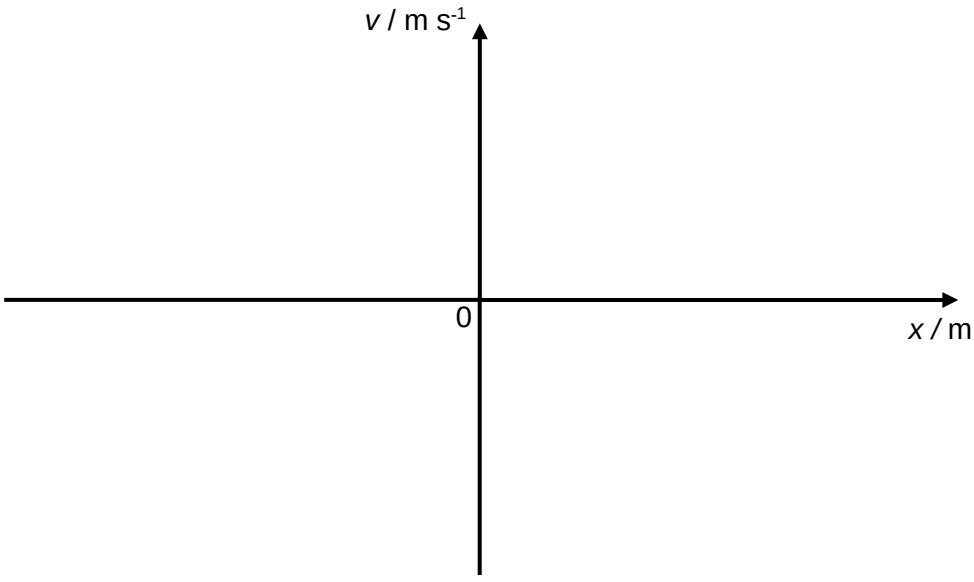


Fig. 4.3

[3]

(b) A student removes the ball and attaches it to the apparatus illustrated in Fig. 4.4

thread

in order to investigate the oscillation vertically.

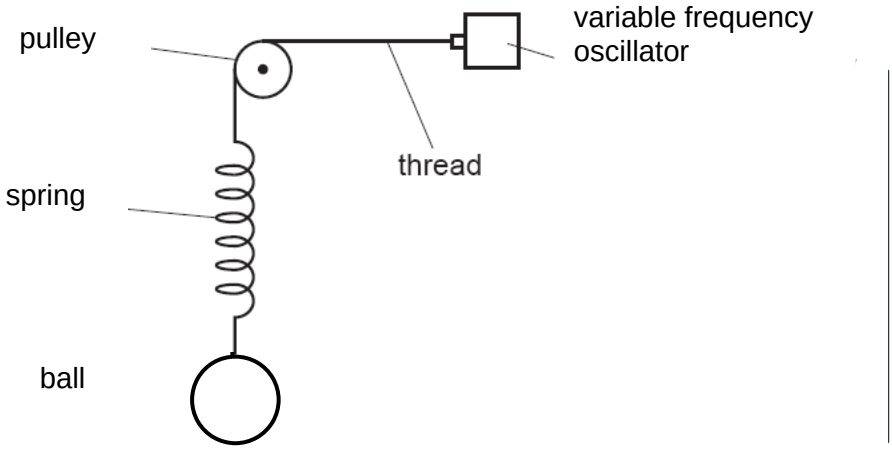
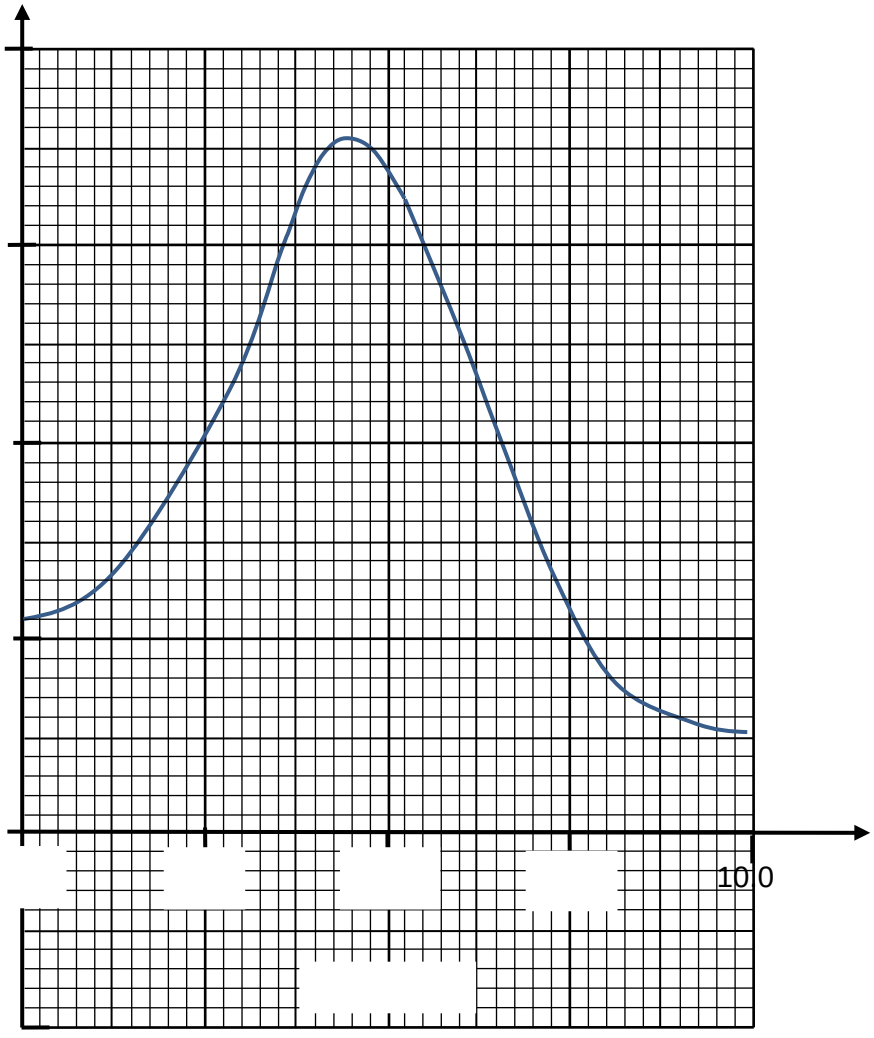


Fig. 4.4

The amplitude of the vibrations produced by the oscillator is constant. The variation with frequency of the amplitude of the oscillations of the ball is shown in Fig. 4.5. The mass of the ball is given to be 150 g, and its oscillations may be assumed to be simple harmonic.



(i) State the amplitude and the frequency when resonance of the system

illustrated in Fig. 4.5 occurs.

amplitude = …… mm

frequency = …… Hz [1]

(ii) **1.** Sketch the free body diagram of the ball when the tension of the spring is at its maximum.

[1]

2. Hence, determine the maximum tension in the spring.

tension = …… N [3]

(iii) Some very light but large feathers are attached to the bottom of the ball so that the feathers extend outwards. The investigation is now repeated.

On Fig. 4.5, draw a line to show the new variation with frequency of the amplitude of variation for frequencies between 2 Hz and 10 Hz. [2]

5 A progressive wave is produced at $x = 0$ on the surface of a liquid. As the wave travels, its amplitude gradually decreases. At one particular time, the variation of the vertical displacement y with distance x along the surface of the liquid is shown in Fig. 5.1.

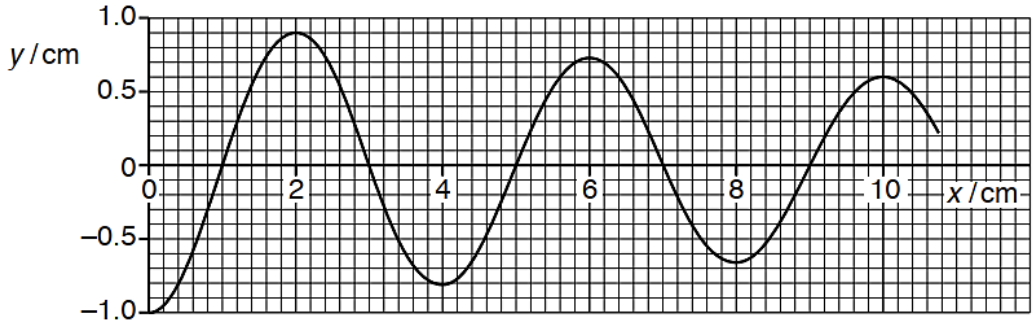


Fig. 5.1

(a) State what is meant by a *progressive wave*.

……

……

…… [1]

(b) From Fig. 5.1, determine the wavelength of the wave

wavelength = …… m [1]

(c) The frequency of the wave is 12.5 Hz. Hence determine the speed of the wave.

speed = …… m s^{-1} [1]

(d) A particle Q is at x = 4.4 cm. Mark and label on Fig. 5.1. the position of a particle P that lags Q in phase by 5.03 rad. Show your workings clearly. [2]

(e) The amplitudes of the first and third crests are shown in Fig. 5.1.

Determine the fraction of the intensity at the third crest to that at the first crest.

fraction = ……. [2]

6 **(a)** State Faraday's law of electromagnetic induction.

……

……

…….. [1]

(b) A solenoid S has a small coil C placed near to one of its ends, as shown in Fig. 6.1.

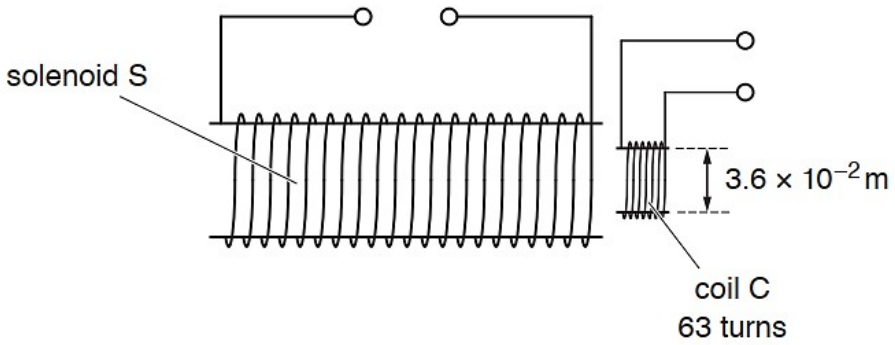


Fig. 6.1

The solenoid S produces a uniform magnetic field of flux density B, in tesla, in the region of coil C given by the expression

$$B = 9.4 \times 10^{-4} I$$

where I is the current, in ampere, in the solenoid S.

The variation with time t of the current I in solenoid S is shown in Fig. 6.2.

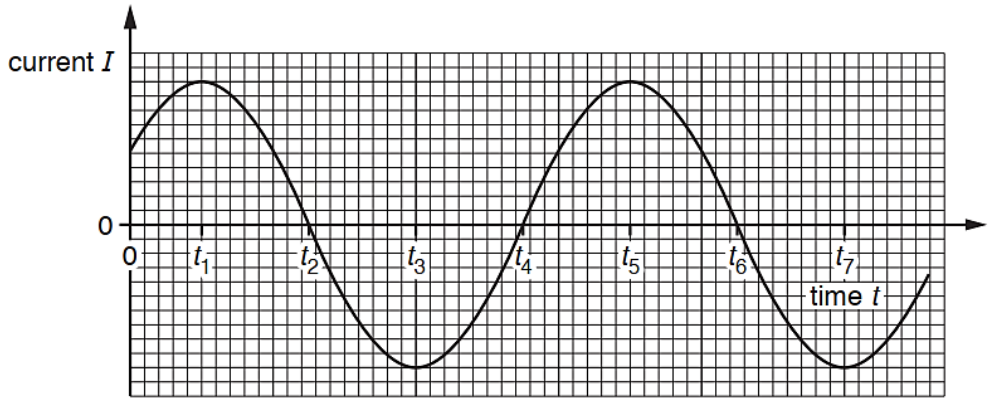


Fig. 6.2

(i) With reference to Fig. 6.2, state two times at which there is no electromotive force (e.m.f.) induced in coil C.

Explain your answer with reference to Fig. 6.2.

time …….. and …….

……

…… [2]

(ii) At a time between t_1 and t_2, state whether the magnetic field produced by the induced current in coil C is in the same direction as or opposite to the magnetic field produced by the solenoid. Explain your answer.

……

……

……

…… [2]

(c) The alternating current in the solenoid S in **(b)** is replaced by a constant current of 5.0 A.

This current is then reversed in a time of 6.0 ms, while maintaining the same magnitude.

Calculate the average e.m.f. induced in coil C during this period.

e.m.f. induced = …… V [3]

7 **(a)** **(i)** Explain how the existence of a threshold frequency in the photoelectric effect provides evidence of the particulate nature of electromagnetic radiation.

……

……

……

…… [2]

(ii) When a beam of electrons is directed at a double slit, an interference pattern is observed on a screen behind the slits. Explain how this observation provides **two** pieces of evidence for the wave nature of electrons.

1 ……

……

2 ……

…… [2]

(b) An electron is confined along the *x*-axis between two rigid walls located at $x = 0$ and $x = 3.00$ cm.

(i) Calculate the energy difference between the ground state and the first excited state of the electron.

energy = …… J [2]

(ii) Hence, determine the wavelength of the photon emitted when the electron transitions from the first excited state to the ground state.

wavelength = …… m [2]

(iii) At a particular instant in time, the electron has the following wavefunction:

$$\psi(x) = \begin{cases} A\sin(kx) & \text{for } 0 \le x \le 3.00\,\text{cm} \\ 0 & \text{elsewhere} \end{cases}$$

where A is the normalisation constant and $k = \dfrac{2\pi}{3.00}$ cm^{-1}.

1. In Fig. 7.1 below, sketch the variation of the probability density $|\psi(x)|^2$ with position x. You are not required to label values.

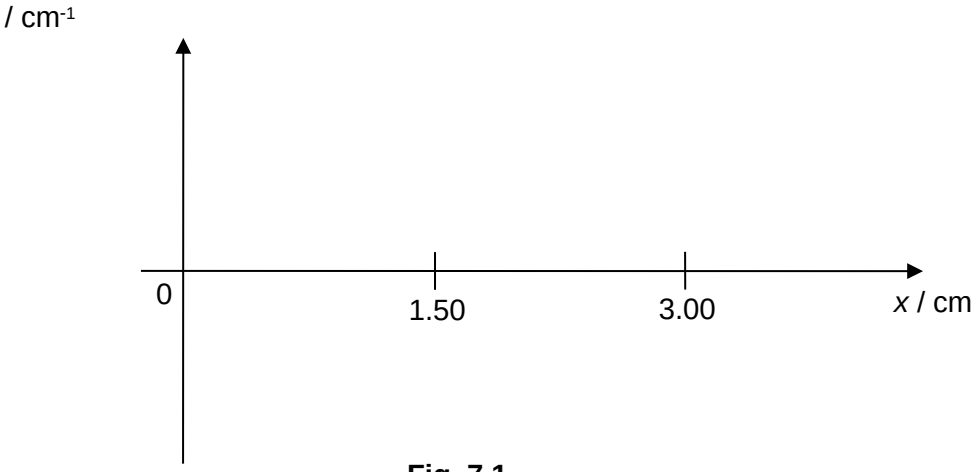


Fig. 7.1

[1]

2. Hence, determine the normalisation constant A.

A = ... cm$^{-\frac{1}{2}}$ [2]

3. Determine the probability of finding the electron between x = 1.50 cm and x = 2.25 cm.

probability = ... [1]

8	The Sun consists largely of hydrogen, $^{1}_{1}\mathrm{H}$ (a proton) and helium, $^{4}_{2}\mathrm{He}$. At its core, temperature reaches a very high value. A continuous process of nuclear fusion occurs,

releasing immense amount of energy. This energy is then radiated by the Sun.

One of the primary reaction pathways responsible for this energy generation is the **proton-proton cycle** (also known as the **hydrogen cycle**). The sequence of nuclear reactions involved in this cycle is detailed below:

$$ {}_{1}^{1}\mathrm{H} + {}_{1}^{1}\mathrm{H} \rightarrow {}_{1}^{2}\mathrm{H} + {}_{+1}^{0}\mathrm{e} + \mathrm{X} $$ ---------- equation (1)

$$ {}_{1}^{2}\mathrm{H} + {}_{1}^{1}\mathrm{H} \rightarrow {}_{2}^{3}\mathrm{He} $$ ---------- equation (2)

$$ {}_{2}^{3}\mathrm{He} + {}_{2}^{3}\mathrm{He} \rightarrow {}_{2}^{4}\mathrm{He} + 2({}_{1}^{1}\mathrm{H}) $$ ---------- equation (3)

In each of these reactions, energy is released. The reactions of this cycle give us most of the energy which we receive from the Sun.

In a nuclear fusion, two small charged nuclei come together to form a larger one. From what we already know about the electric potential energy between charged particles, we can work out the conditions necessary for the fusion. From the kinetic theory of gases, the temperature *T* of the gas and the mean kinetic energy $\mathrm{E_k}$ of the particles of the gas like ${}_{1}^{1}\mathrm{H}$ is given by:

$$ \mathrm{E_k} = \frac{3}{2}\mathrm{kT} $$

where *k* is the Boltzmann's constant.

The Sun radiates energy into space at a constant rate of 3.8×10^{26} W. This power output is sustained entirely by nuclear fusion reactions occurring within the solar core, where protons fuse to form helium.

The Sun has a total mass of 2.0×10^{30} kg. However, the high temperature required for nuclear fusion exist only in its core. As a result, the hydrogen (protons) available for fusion in the core accounts for only 10% of the Sun's total mass.

In the core's fusion cycle, the average net energy released *per proton* consumed is 6.68 MeV.

You may use relevant information in Fig. 8.1 to answer the following questions

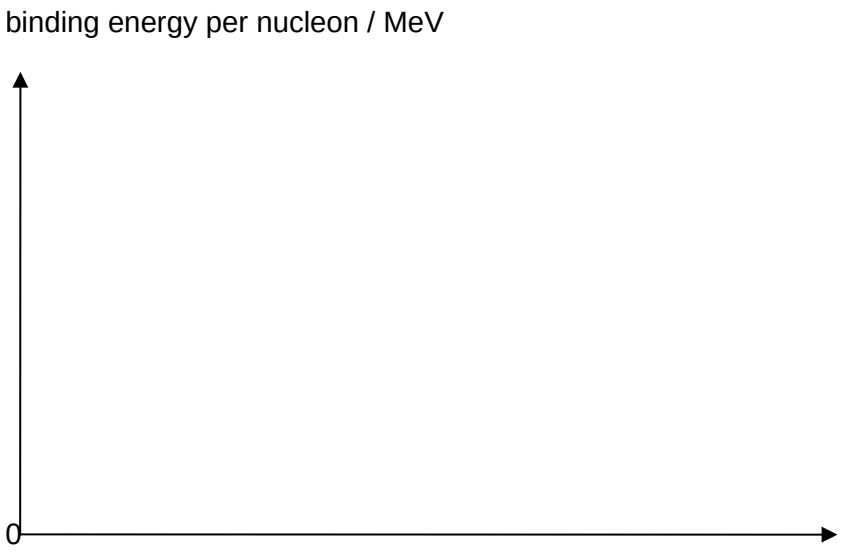
Nuclide	Mass / *u*
$^{0}_{1}e$	0.000 549
$^{1}_{1}H$	1.007 825
$^{1}_{0}n$	1.008 665
$^{2}_{1}H$	2.014 102
$^{3}_{1}H$	3.016 049
$^{3}_{2}He$	3.016 030
$^{4}_{2}He$	4.002 604

Fig. 8.1

(a) **(i)** Determine the binding energy per nucleon of the helium $^{4}_{2}He$ nuclide.

binding energy per nucleon = …… MeV [2]

(ii) Sketch the binding energy per nucleon with for nucleon number from A = 1 to A = 240.



[1]

(iii) Use your answer in **(a)(i)** and **(a)(ii)** to explain why nuclear fusion may release energy.

……

…… [1]

(iv) Show that the energy released for equation (2) is 5.5 MeV.

[1]

(b) In the first reaction, shown in equation (1), it is observed that the particle $^{0}_{+1}e$, a positron, is emitted with varying amounts of kinetic energy.

Use relevant conservation law(s) to suggest why emission of a positron in

equation (1) could predict the existence of another particle X.

……

……

……

……

…… [2]

(c) As one proton approaches the other, electrostatic repulsion causes it to slow down, converting their kinetic energy into electric potential energy. For nuclear fusion to take place, the protons must possess enough kinetic energy to reach a separation of 1.0×10^{-14} m where the attractive strong nuclear force becomes dominant.

To fuse two protons together in this way, the temperature must be raised. Using kinetic theory of gases, estimate the minimum temperature necessary for protons to achieve this fusion condition.

temperature = …… K [2]

(d) With reference to equation (1), (2) and (3), determine the net number of protons that are required to produce a helium $^{4}_{2}\mathrm{He}$ nucleus.

number = …… [1]

(e) **(i)** Show that there are 1.2×10^{56} protons available for nuclear fusion in the Sun's core.

[1]

(ii) Use **(e)(i)** to estimate the lifetime of the Sun. Give your answer in years.

lifetime of the Sun = …… years [1]

END OF PAPER